

Mini-Project 3

Group E

Aya Jroundi (Sciper: 341943)

Raphael Javary (Sciper: 342263)

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1 Introduction

This mini-project involves the dynamic analysis of a simply supported square plate using finite element methods. The plate is defined by four edges simply supported and is modeled with two types of bending elements over structured meshes. Key parameters of the plate are:

- Side length: $\ell = 0.5$ m
- Thickness: $h = 0.005$ m
- Young's modulus: $E = 210$ GPa
- Poisson's ratio: $\nu = 0.3$
- Density: $\rho = 7850$ kg/m³

Project Objectives:

- Compute the first fundamental natural frequencies and corresponding mode shapes for meshes of 2×2 , 4×4 , 8×8 , and 16×16 elements using AMC and CR plate bending elements.
- Compare numerical frequencies with exact analytical solutions to assess convergence behavior.
- Investigate the effect of span-to-thickness ratio (ℓ/h from 100 to 10) on the accuracy of the lowest frequency for both thin and thick plate formulations.

Software and Approach: All analyses are performed in MATLAB by assembling global stiffness and mass matrices, followed by eigenvalue extraction to obtain natural frequencies and mode shapes.

2 Methodology

2.1 Assumptions

The following assumptions are made for the finite element model:

- The plate is thin ($h \ll \ell$), so Kirchhoff–Love theory applies (no transverse shear deformation).
- Material is homogeneous, isotropic, and linear elastic (constant E, ν).
- Displacements and rotations are small (geometrically linear analysis).
- No damping or external loads other than initial free vibration (mass and stiffness only).
- All four edges are simply supported (pinned): transverse displacement $u_3 = 0$ and bending moment $M\{u_3\} = 0$ on $\partial\Omega$. On horizontal edges, the rotation of nodes in y is blocked (as this would violate the non displacement condition for neighboring nodes): $\partial_{\epsilon_2} u_3 = 0$. Analogously on vertical edges $\partial_{\epsilon_1} u_3 = 0$. For CR elements the twist $\partial_{\epsilon_1 \epsilon_2}^2 u_3$ is free on all edges.
- Self-weight and in-plane forces are neglected.

2.2 Model Setup

Justification of the Thin-Plate Model : The Kirchhoff–Love theory is valid in the thin-plate limit, typically for a span-to-thickness ratio $\ell/h > 10$ –15. Beyond this threshold, transverse-shear deformations are no longer negligible, and the Timoshenko plate theory (which includes a shear-deformation term) should be employed.

Let $\Omega = [0, \ell] \times [0, \ell]$ denote the midplane of the square plate. We seek the transverse displacement $u_3(x, t) \in C^4(\Omega \times [0, T])$ satisfying the strong form of Kirchhoff–Love plate bending:

$$\frac{h^3}{12} \nabla_k^T \mathbf{C} \nabla_k u_3 + \rho h \ddot{u}_3 = f_3 \quad \text{on } \Omega \times (0, T),$$

with simply supported boundary conditions:

$$\nabla_k = (\partial_{x_1}^2, \partial_{x_2}^2, 2\partial_{x_1 x_2}^2), \quad u_3 = 0, \quad M\{u_3\} = 0 \quad \text{on } \partial\Omega,$$

and initial conditions $u_3(\cdot, 0) = u_0, \dot{u}_3(\cdot, 0) = v_0$ in Ω .

From Strong to Weak Form. We consider the transverse displacement $u_3(x, t)$ on the square mid-plane $\Omega = [0, \ell] \times [0, \ell]$, satisfying

$$\frac{h^3}{12} \nabla_k^T (C \nabla_k u_3) + \rho h \ddot{u}_3 = f_3 \quad \text{in } \Omega \times (0, T), \quad (1)$$

with simply supported boundary conditions

$$u_3 = 0, \quad M\{u_3\} = 0 \quad \text{on } \partial\Omega, \quad (2)$$

and initial conditions $u_3(\cdot, 0) = u_0, \dot{u}_3(\cdot, 0) = v_0$.

Here

$$\nabla_k = \begin{pmatrix} \partial_{x_1}^2 \\ \partial_{x_2}^2 \\ 2\partial_{x_1 x_2} \end{pmatrix},$$

and $M\{u_3\} = n \cdot (C \nabla_k u_3)$ is the boundary bending moment.

Multiply by a Virtual Displacement Let $\delta u_3 \in V$ be a test-function. Multiply (1) by δu_3 and integrate over Ω :

$$\int_{\Omega} \delta u_3 \left[\frac{h^3}{12} \nabla_k^T (C \nabla_k u_3) + \rho h \ddot{u}_3 - f_3 \right] d\Omega = 0. \quad (3)$$

Split into

$$I_1 = \frac{h^3}{12} \int_{\Omega} \delta u_3 \nabla_k^T (C \nabla_k u_3) d\Omega, \quad I_2 = \rho h \int_{\Omega} \delta u_3 \ddot{u}_3 d\Omega, \quad I_3 = - \int_{\Omega} \delta u_3 f_3 d\Omega.$$

Integration by Parts of the Bending Term : Focus on

$$I_1 = \frac{h^3}{12} \int_{\Omega} \delta u_3 \nabla_k^T (C \nabla_k u_3) d\Omega.$$

Since $\nabla_k u_3$ contains second derivatives, we apply integration by parts twice.

First integration by parts : Set $v = C \nabla_k u_3$. Then

$$\int_{\Omega} \delta u_3 \nabla_k^T v d\Omega = \int_{\partial\Omega} \delta u_3 (n \cdot v) ds - \int_{\Omega} (\nabla_k \delta u_3)^T v d\Omega,$$

where n is the outward normal. The boundary term is $\delta u_3 M\{u_3\}$.

Second integration by parts : The term $\int_{\Omega} (\nabla_k \delta u_3)^T (C \nabla_k u_3) d\Omega$ still has second derivatives in its factors. Integrating by parts again in each component transfers one derivative onto δu_3 , producing

$$\int_{\Omega} (\nabla_k \delta u_3)^T (C \nabla_k u_3) d\Omega = - \int_{\Omega} \delta u_3 \nabla_k^T (C \nabla_k u_3) d\Omega + \int_{\partial\Omega} (\partial_n \delta u_3) Q\{u_3\} ds.$$

Collecting gives

$$I_1 = \frac{h^3}{12} \left[- \int_{\Omega} (\nabla_k \delta u_3)^T C (\nabla_k u_3) d\Omega + \int_{\partial\Omega} (\delta u_3 M\{u_3\} - (\partial_n \delta u_3) Q\{u_3\}) ds \right].$$

Apply Simply-Supported BCs : On $\partial\Omega$, $u_3 = 0 \implies \delta u_3 = 0$, and $M\{u_3\} = 0$, plus $\partial_n u_3 = 0$ kills the shear-term. Hence all boundary integrals vanish, leaving

$$I_1 = - \frac{h^3}{12} \int_{\Omega} (\nabla_k \delta u_3)^T C (\nabla_k u_3) d\Omega.$$

Assemble the Weak Form : Plugging I_1, I_2, I_3 into (3), moving the minus-sign:

$$\frac{h^3}{12} \int_{\Omega} (\nabla_k \delta u_3)^T C (\nabla_k u_3) d\Omega + \rho h \int_{\Omega} \delta u_3 \ddot{u}_3 d\Omega = \int_{\Omega} f_3 \delta u_3 d\Omega,$$

for all $\delta u_3 \in V$, where

$$U = V = \{v \in H^2(\Omega) \mid v|_{\partial\Omega} = 0, \partial_n v|_{\partial\Omega} = 0\}.$$

Summary

1. Multiply the PDE by δu_3 and integrate over Ω .
2. Integrate by parts twice to shift derivatives onto the test function.
3. Use boundary conditions to kill all boundary terms.
4. Identify the bilinear form and mass term on $H^2(\Omega)$.

Multiplying by a virtual displacement $\delta u_3 \in V$ and integrating by parts yields the weak (variational) form: find $u_3 \in U$ such that for all $\delta u_3 \in V$,

$$\int_{\Omega} \frac{h^3}{12} \nabla_k \delta u_3^T \mathbf{C} \nabla_k u_3 d\Omega + \int_{\Omega} \rho h \delta u_3 \ddot{u}_3 d\Omega = \int_{\Omega} f_3 \delta u_3 d\Omega.$$

We define the trial and test spaces as

$$U = \{v \in H^2(\Omega) \mid v|_{\partial\Omega} = 0, \partial_n v|_{\partial\Omega} = 0\}, \quad V = U.$$

2.3 Element Kinematics and Interpolation

The plate is discretized with two bending element formulations over a structured $n \times n$ mesh: nonconforming AMC elements and conforming CR elements. Each element has 12 DOFs: the transverse displacement and two rotations (θ_x, θ_y) at each node.

Coordinate Transformation. Each element is mapped from the reference domain $\hat{\Omega} = [-1, 1]^2$ to a physical element of size $a = \ell/n$ by the affine mapping

$$x(\boldsymbol{\xi}) = a\boldsymbol{\xi}, \quad \boldsymbol{\xi} = (\xi_1, \xi_2)^T.$$

AMC Approximation. Within each AMC element, u_3 is approximated by a complete cubic polynomial in (ξ_1, ξ_2) augmented with two quartic invariants (12 basis functions). Continuity of u_3 is enforced, but C^1 slope continuity is not (nonconforming). With this choice of basis functions, each element has 3 degrees of freedom: its vertical displacement u_3 , as well as rotations in x and y: $\partial_{e_2} u_3$ and $\partial_{e_1} u_3$.

CR Approximation. The CR element enforces both u_3 and slope continuity (C^1) by employing quartic polynomial shape functions satisfying the strong form interpolation conditions. Consequently, there are 16 local shape functions and four degrees of freedom per node, with the extra one corresponding to the twist: $\partial_{e_1 e_2}^2 u_3$

Strain–Displacement Matrix. The curvature–displacement relation is given by $\mathbf{B} = \nabla_k \mathbf{H}$, where $\mathbf{H}$ collects the 12 shape functions in local coordinates:

$$\mathbf{B}(\boldsymbol{\xi}) = \begin{bmatrix} \frac{1}{a^2} \partial_{\xi_1}^2 \\ \frac{1}{a^2} \partial_{\xi_2}^2 \\ \frac{2}{a^2} \partial_{\xi_1 \xi_2}^2 \end{bmatrix} \mathbf{H}(\boldsymbol{\xi}).$$

Element Matrices and Loads. Element contributions are computed as

$$\mathbf{K}^e = \frac{h^3 a^2}{12} \int_{-1}^1 \int_{-1}^1 \mathbf{B}^T \mathbf{C} \mathbf{B} d\xi_1 d\xi_2, \quad \mathbf{M}^e = \rho h a^2 \int_{-1}^1 \int_{-1}^1 \mathbf{H}^T \mathbf{H} d\xi_1 d\xi_2,$$

$$\mathbf{r}^e(t) = a^2 \int_{-1}^1 \int_{-1}^1 \mathbf{H}^T f_3(\boldsymbol{\xi}, t) d\xi_1 d\xi_2.$$

Global assembly and eigenvalue analysis yield natural frequencies and mode shapes.

Global Assembly of K and M in the AMC Case: We number each node $i = 1, \dots, n$ and collect the four corner node indices of element e in a connectivity array

$$\text{connectivity}(e, :) = [i_1, i_2, i_3, i_4].$$

Each global node has three DOFs $(u_3, \theta_x, \theta_y)$, so global matrices $K, M \in \mathbb{R}^{3n \times 3n}$ are initialized to zero. Then, for each element $e = 1, \dots, n_e$ and each pair of local nodes $a, b = 1, \dots, 4$, we add the 3×3 blocks K_{ab}^e and M_{ab}^e into the global matrices at the corresponding global-DOF positions:

$$K[3i_a - 2 : 3i_a, 3i_b - 2 : 3i_b] += K_{ab}^e, \quad M[3i_a - 2 : 3i_a, 3i_b - 2 : 3i_b] += M_{ab}^e,$$

where $i_a = \text{connectivity}(e, a)$ and $i_b = \text{connectivity}(e, b)$.

Applying Boundary Conditions: The global degrees of freedom corresponding to displacement on edge nodes are identified and corresponding rows and columns deleted. On horizontal edges, the rotation in x degree of freedom is deleted. On vertical edges, rotation in y is blocked. For CR elements which have an additional twist degree of freedom per node, it is free on all edges.

2.4 Analysis Type

In this mini-project, we perform a finite-element modal analysis of a simply supported square plate to investigate its dynamic behavior. Two bending element formulations are compared: the nonconforming Affine Macro-Cell (AMC) element and the conforming Rectangular (CR) element. After generating structured meshes of varying density, we compute the first and second natural frequencies and corresponding mode shapes for each formulation.

We also study the effect of plate thickness (span-to-thickness ratio ℓ/h) on mode shapes and frequencies to highlight the importance of transverse-shear corrections in thick-plate theory.

Algorithm Outline : A pseudocode or flowchart can be added here, for instance:

1. Generate structured $n \times n$ mesh.
2. Compute element matrices K^e, M^e .
3. Assemble global matrices K, M .
4. Apply boundary conditions.
5. Solve eigenvalue problem.
6. Post-process and plot mode shapes.

3 Results and Discussion

3.1 Meshing

Structured meshes of sizes 2×2 , 4×4 , 8×8 , and 16×16 were generated for both AMC and CR plate elements. Figure 1 illustrates representative meshes.

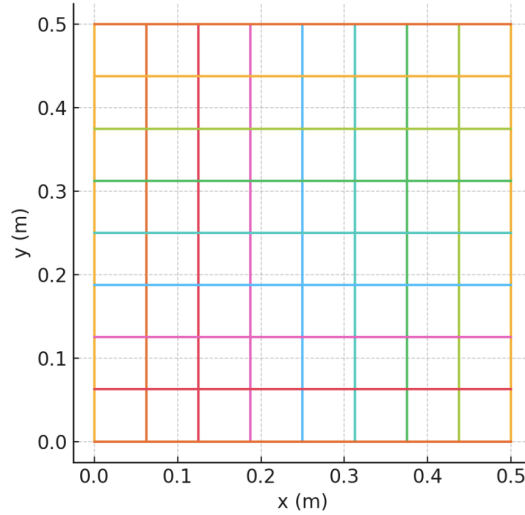


Figure 1: 8x8 Mesh Plate

3.2 Frequency Convergence

Exact solutions are known in the case of a simply-supported flat plate. Analytical natural frequencies for a thin plate are given by: $\omega_{m,n} = \pi^2 \frac{m^2+n^2}{l^2} \sqrt{\frac{D}{\rho h}}$, where $D = \frac{Eh^3}{12(1-\nu^2)}$.

Table 1 is prepared to compare the computed first natural frequency ω_1 against the analytical $\omega_{1,1}$:

Mesh size	ω_{analytic}	ω_{AMC}	ω_{CR}	Error AMC (%)	Error CR (%)
2×2	617.9	564.1	619.2	-8.7	0.22
4×4	617.9	599.4	618.0	-2.9	1.33×10^{-2}
8×8	617.9	612.8	617.9	-0.8	8.28×10^{-4}
16×16	617.9	616.6	617.9	-0.2	5.16×10^{-5}

Table 1: First fundamental frequencies and convergence behavior.

Both elements converge toward the exact value with finer meshes.

Table 2 compares the computed second natural frequency ω_2 ; which is equal to the third natural frequency; against the analytical $\omega_{1,2}$:

Mesh size	ω_{analytic}	ω_{AMC}	ω_{CR}	Error AMC (%)	Error CR (%)
2×2	1544.7	1593.7	1658.8	3.1	7.4
4×4	1544.7	1483.9	1548.8	-3.9	0.26
8×8	1544.7	1525.7	1545.0	-1.2	1.68×10^{-2}
16×16	1544.7	1544.8	1553.4	-0.3	1.06×10^{-3}

Table 2: Third natural frequencies and convergence behavior.

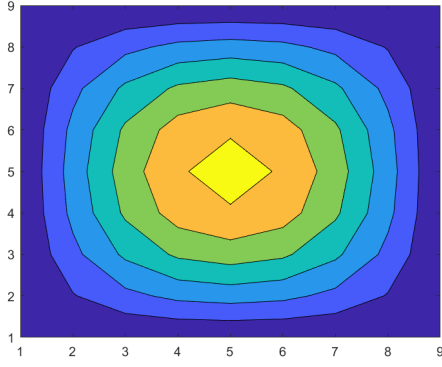
And eventually, Table 3 compares the computed fourth natural frequency ω_4 against the analytical $\omega_{2,2}$:

Mesh size	ω_{analytic}	ω_{AMC}	ω_{CR}	Error AMC (%)	Error CR (%)
2×2	2471.6	3398.4	2626.5	37	6.3
4×4	2471.6	2256.5	2477.0	-8.7	0.22
8×8	2471.6	2397.8	2472.0	-2.9	1.33×10^{-2}
16×16	2471.6	2471.6	2493.3	-0.8	8.28×10^{-4}

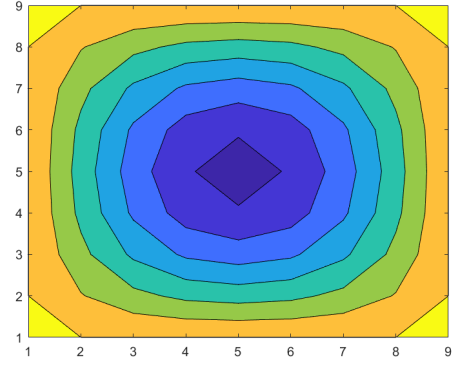
Table 3: Fourth natural frequencies and convergence behavior.

3.3 First, Second and Fourth Mode Shapes for AMC vs. CR

To compare the basic dynamic behavior, Figures 2, 3 and Figure 4 show the first, second and fourth mode shapes, respectively, for both AMC and CR elements on an 8×8 mesh.

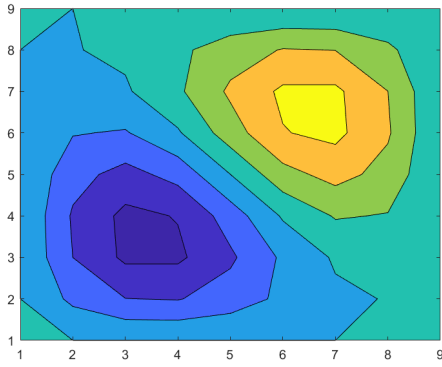


(a) First mode for AMC plate

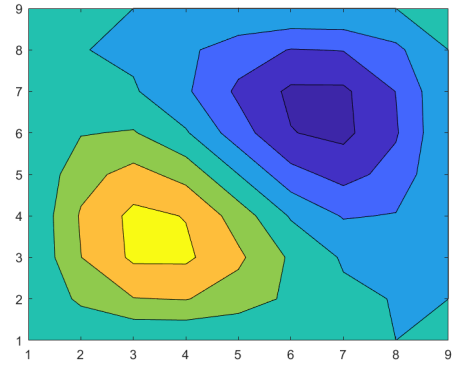


(b) First mode for CR plate

Figure 2: Comparison of first (fundamental) mode shapes for AMC and CR elements.

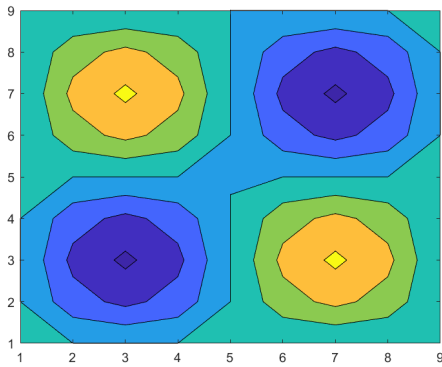


(a) Second mode for AMC plate.

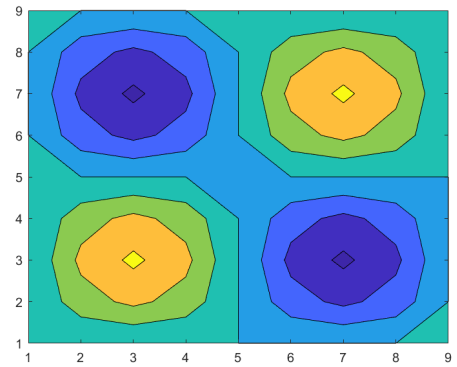


(b) Second mode for CR plate.

Figure 3: Comparison of second mode shapes for AMC and CR elements.



(a) Fourth mode for AMC plate



(b) Fourth mode for CR plate

Figure 4: Comparison of fourth mode shapes for AMC and CR elements.

Both elements can be seen effectively recreating the expected mode shapes. It is interesting to note that the nodal line in mode 2 (identical to mode 3, just rotated 90 degrees) is not exactly diagonal. Depending on the plate's boundary conditions and thickness to

span ratio, this mode can exhibit any nodal line angle between diagonal and fully vertical/horizontal [1]. Additionally, the CR element’s C^1 continuity produces smoother curvature fields in both the fundamental and higher modes, while the AMC element may show minor slope discontinuities at element interfaces.

3.4 Effect of Span-to-Thickness Ratio

To study shear effects, Table 4 outlines first natural frequency variation versus span-to-thickness ratio ℓ/h , comparing thin-plate (Kirchhoff–Love) and thick-plate formulations. Thin plate frequencies can be obtained equivalently through the analytical formula or a FEM simulation (16x16 with CR elements, for which error is practically zero). Notice that plate stiffness scales with h^3 while plate mass scales with h , as such natural frequencies simply scale with h under the thin plate model. For thick plates, an analytical formula for the first mode also exists, given by: $\omega_{1,1}(h) = 20\pi^2 h \sqrt{\frac{70}{4\pi^2 h^2 + 7}}$.

ℓ/h	ω_{thin}	ω_{thick}	Error (%)
100	617.9	617.7	0.02
50	1235.8	1234.4	0.1
20	3089.5	3068.0	0.6
10	6179.1	6011.9	2.7

Table 4: Frequency variation with span-to-thickness ratio.

As the span-to-thickness ratio ℓ/h increases (i.e. the plate becomes thinner), shear-deformation effects become negligible and the thin-plate (Kirchhoff–Love) and thick-plate predictions converge (the error falls to just 0.02 % at $\ell/h = 100$). Conversely, for smaller ℓ/h (thicker plates), shear deformation significantly reduces the natural frequency predicted by the thin-plate model, leading to a larger discrepancy (2.7% error at $\ell/h = 10$).

3.5 Effect of Plate Thickness on Mode Shapes

Mode shapes can also change noticeably if the plate’s thickness varies. In particular, the second mode is known to shift to a more vertical nodal line as plate thickness is increased [1].

Figure 6 illustrates the second mode for a “thick” plate ($\ell/h = 10$), for each element.

As thickness increases (lower ℓ/h), no visible effect can be seen on the modal shapes, suggesting that both CR and AMC elements are unable to capture the experimental dynamics of plate thickening. This is unsurprising since the FEM analysis is based on Kirchhoff–Love theory which neglects the effect of shear deformation, which plays a major role in thicker plates. To see a difference in behavior with thickening, we would need to incorporate shear into the model.

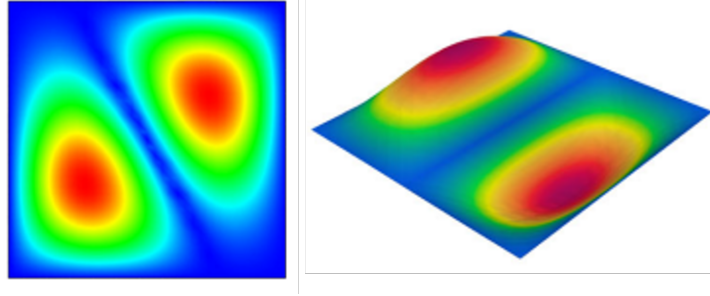


Figure 5: Vertical versus semi-diagonal mode (2,1) shapes, observed depending on plate thickness and clamp/pinned boundary conditions [1]

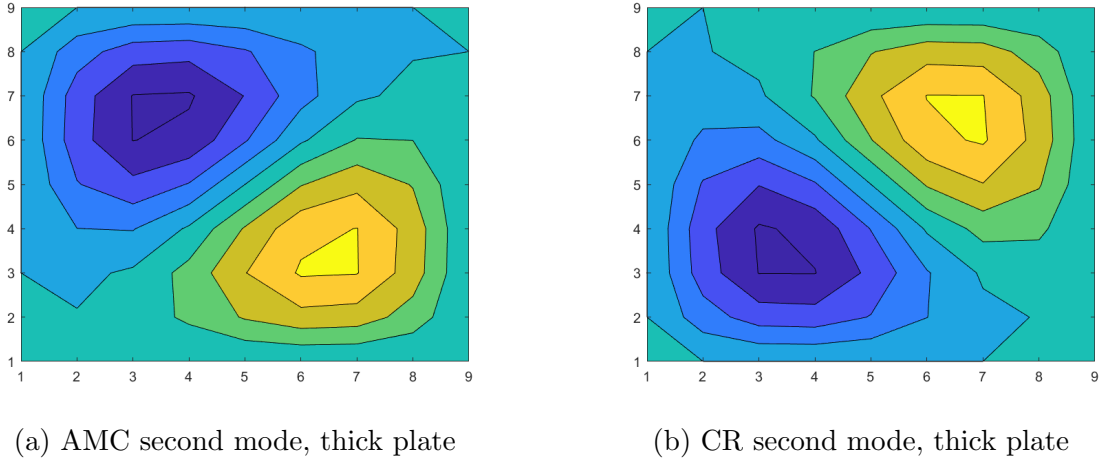


Figure 6: AMC and CR second mode shapes for thick ($\ell/h = 10$) plates.

3.6 Interpretation and Comparison

From tables 1-3, it can be seen that both AMC and CR natural elements produce natural frequencies which converge to their analytical value as meshes are refined. AMC elements can be observed converging at an order of 2 (error is divided by 4 when element size is divided by 2). CR elements converge much more rapidly at an order of 4 (error is divided by 16 when element size is divided by 2). Thus, while error remains at the order of 0.1% for the fundamental frequency at the finest mesh for AMC elements, it practically vanishes for its CR counterpart. This increased accuracy comes at the price of a larger number of degrees of freedom for CR elements (increased by $4/3$) with an accompanied increase in computation time. Still, an CR element achieves a similar error in natural frequency for a 2×2 element as an AMC element in a 16×16 mesh, which obviously requires much less computational power despite the added twist degree of freedom. Thus, in order to achieve similar errors, CR elements allow usage of coarser meshes and less computational time, thus clearly dominate their AMC counterparts in usefulness.

In terms of modal shape, the absence of easily implementable analytical solution (especially for mode 2) makes it difficult to validate numerical results. Nevertheless, both elements produce visually indistinguishable displacement field. The notable difference comes from the fact that AMC elements are nonconforming, while CR elements ensure $C1$ continuity, thus producing a more realistic description of the mode shape. Nonetheless, when looking at displacements only, both element types perform very well.

Another interesting observation is that AMC elements almost systematically underestimate natural frequencies, whereas CR elements overestimate natural frequency (a result which is theoretically guaranteed) as is the case for FEM approximations of bars and beams.

4 Conclusion

This investigation has shown that both AMC and CR elements are able to effectively capture the natural frequencies and modes of a flat square plate, with errors converging below 1% with relatively few elements (8 per dimension). Error comparison has shown that CR elements clearly outperform their AMC counterparts and are able to achieve similar errors with drastically coarser meshes. Coupled with their better continuous approximation of mode shapes, CR elements are identified as the element of choice for Kirchhoff-Love plate theory calculations. Additionally, AMC elements have been found to regularly underestimate natural frequencies, while CR elements are systematically over-stiff.

The investigation has not shown behavioral changes with increasing plate thickness, aside from plate stiffness scaling with the thickness cubed. This is because analysis has been based on the thin plate Kirchhoff-Love strong form. By extending the Kirchhoff-Love model into the domain of thicker plates, significant errors are made with regards to the modal shapes and frequencies. The transition to a modal shape with a vertical nodal line has not been observed, and natural frequencies were overestimated by 2.7% even with a refined 16x16 meshed. It can be said that the FEM predictions developed in this project cannot be reliably extended to the thick plate range ($l/h < 50$). Incorporating Reissner-Mindlin plate theory is necessary to capture the effects of shear deformation in these thick plates.

References

1. Pagani, A., Azzara, R. and Carrera, E. "Geometrically nonlinear analysis and vibration of in-plane-loaded variable angle tow composite plates and shells". *Acta Mech* 234, 85–108 (2023). <https://doi.org/10.1007/s00707-022-03226-2>